A BASIS OF B AS AN A-MODULE VIA T DESCENDS TO A BASIS OF A OVER THE INVARIANTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma lemma-diagram-pull (canonical tag 03C6). cited at proof lines 4980-4981 and 5007-5009. Produces the cocartesian squares and the commutativity c(t(x_i)) = p_0(t(x_i)).
- Lemma descent-lemma-ff-exact (canonical tag 023M). cited at proof lines 4996-5000. The top row is an equalizer diagram since s is faithfully flat.
- Lemma lemma-diagram (canonical tag 02YE) and equation equation-invariants (03BF). cited at proof lines 4981 and 5003. Commutativity of the top part and the lower equalizer row by definition of C.

Source reference. Original lemma 03C6: sources/stacks-chow/tags.txt, source lines 4373–4373. Cited at proof lines 4980-4981 to produce the commutative diagram with cocartesian squares and the commutativity used at line 5007; load-bearing.

Source reference. Original lemma 023M: sources/stacks-chow/tags.txt, source lines 2747–2747. Cited at proof lines 4996-5000: condition (2) makes s faithfully flat, so the top row is an equalizer diagram; load-bearing.

Source reference. Original lemma 02YE: sources/stacks-chow/tags.txt, source lines 3873–3873. Cited at proof lines 5007-5009 for the equality c(t(x_i)) = p_0(t(x_i)); load-bearing.

Source reference. Original equation 03BF: sources/stacks-chow/tags.txt, source lines 4346–4346. Cited at proof line 4981 (definition of C) and line 5003 (lower row an equalizer by definition of C); load-bearing.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume*

(1) *$U = \mathrm{Spec}(A)$, and $R = \mathrm{Spec}(B)$ are affine, and*
(2) *there exist elements $x_i \in A$, $i \in I$ such that $B = \bigoplus_{i \in I} s^\sharp(A)t^\sharp(x_i)$.*

Then $A = \bigoplus_{i \in I} Cx_i$, and $B \cong A \otimes_C A$ where $C \subset A$ is the R-invariant functions on U as in (03BF).

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Proof. During this proof we will write $s, t : A \to B$ instead of $s^\sharp, t^\sharp$, and similarly $c : B \to B \otimes_{s,A,t} B$. We write $p_0 : B \to B \otimes_{s,A,t} B$, $b \mapsto b \otimes 1$ and $p_1 : B \to B \otimes_{s,A,t} B$, $b \mapsto 1 \otimes b$. By Lemma 03C6 and the definition of C we have the following commutative diagram

$$\begin{array}{ccccc} B \otimes_{s,A,t} B & \overset{c}{\underset{p_0}{\leftleftarrows}} & B & \xleftarrow[t]{} & A \\ \uparrow{\scriptstyle p_1} & & \uparrow{\scriptstyle s} & & \uparrow \\ B & \overset{s}{\underset{t}{\leftleftarrows}} & A & \longleftarrow & C \end{array}$$

Moreover the tow left squares are cocartesian in the category of rings, and the top row is isomorphic to the diagram

$$B \otimes_{t,A,t} B \overset{p_1}{\underset{p_0}{\leftleftarrows}} B \xleftarrow[t]{} A$$

which is an equalizer diagram according to Descent, Lemma 023M because condition (2) implies in particular that s (and hence also then isomorphic arrow t) is faithfully flat. The lower row is an equalizer diagram by definition of C. We can use the x_i and get a commutative diagram

$$\begin{array}{ccccc} B \otimes_{s,A,t} B & \overset{c}{\underset{p_0}{\leftleftarrows}} & B & \xleftarrow[t]{} & A \\ \uparrow{\scriptstyle p_1} & & \uparrow{\scriptstyle s} & & \uparrow \\ \bigoplus_{i\in I} Bx_i & \overset{s}{\underset{t}{\leftleftarrows}} & \bigoplus_{i\in I} Ax_i & \longleftarrow & \bigoplus_{i\in I} Cx_i \end{array}$$

where in the right vertical arrow we map x_i to x_i, in the middle vertical arrow we map x_i to $t(x_i)$ and in the left vertical arrow we map x_i to $c(t(x_i)) = t(x_i) \otimes 1 = p_0(t(x_i))$ (equality by the commutativity of the top part of the diagram in Lemma 02YE). Then the diagram commutes. Moreover the middle vertical arrow is an isomorphism by assumption. Since the left two squares are cocartesian we conclude that also the left vertical arrow is an isomorphism. On the other hand, the horizontal rows are exact (i.e., they are equalizers). Hence we conclude that also the right vertical arrow is an isomorphism. □